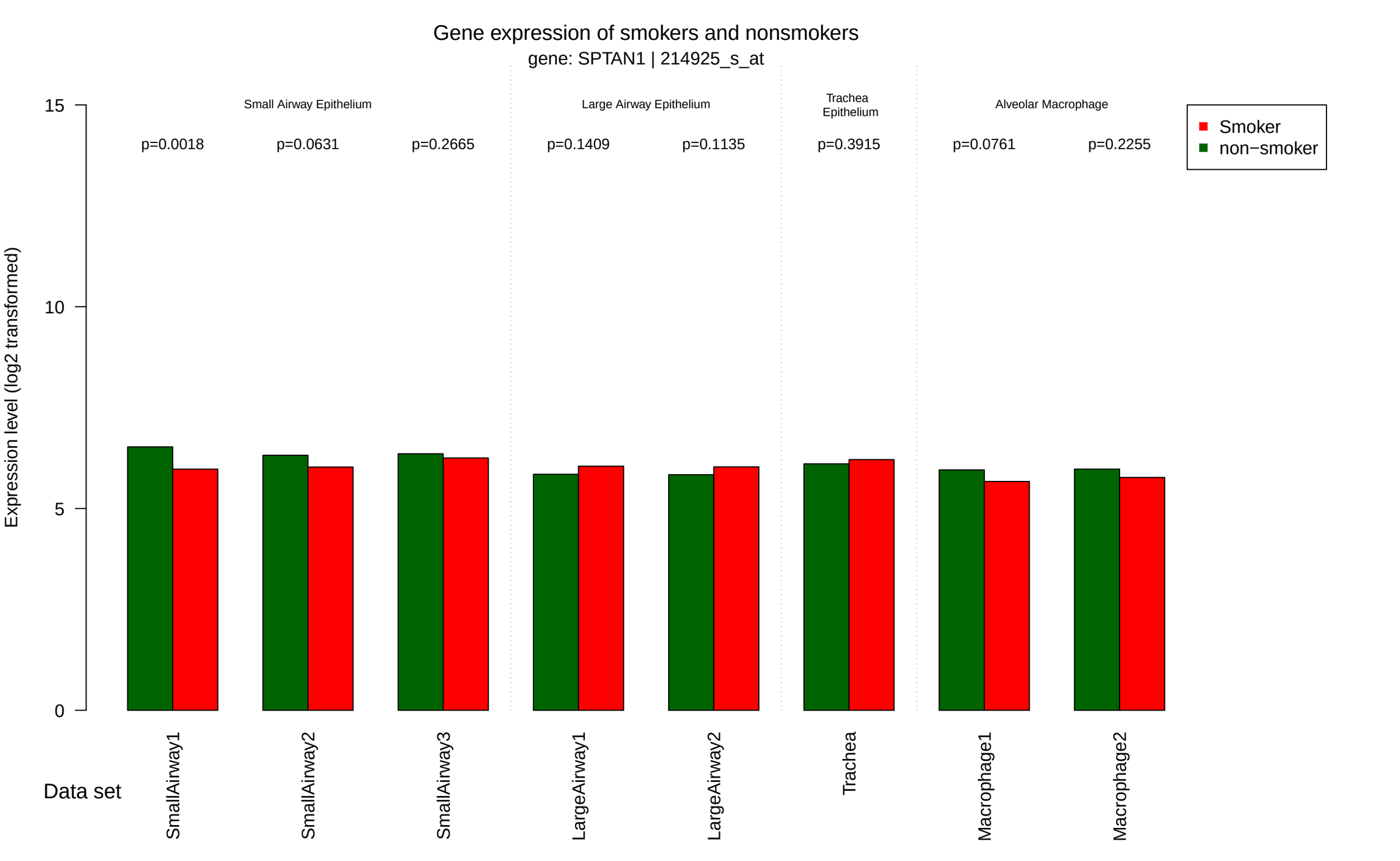
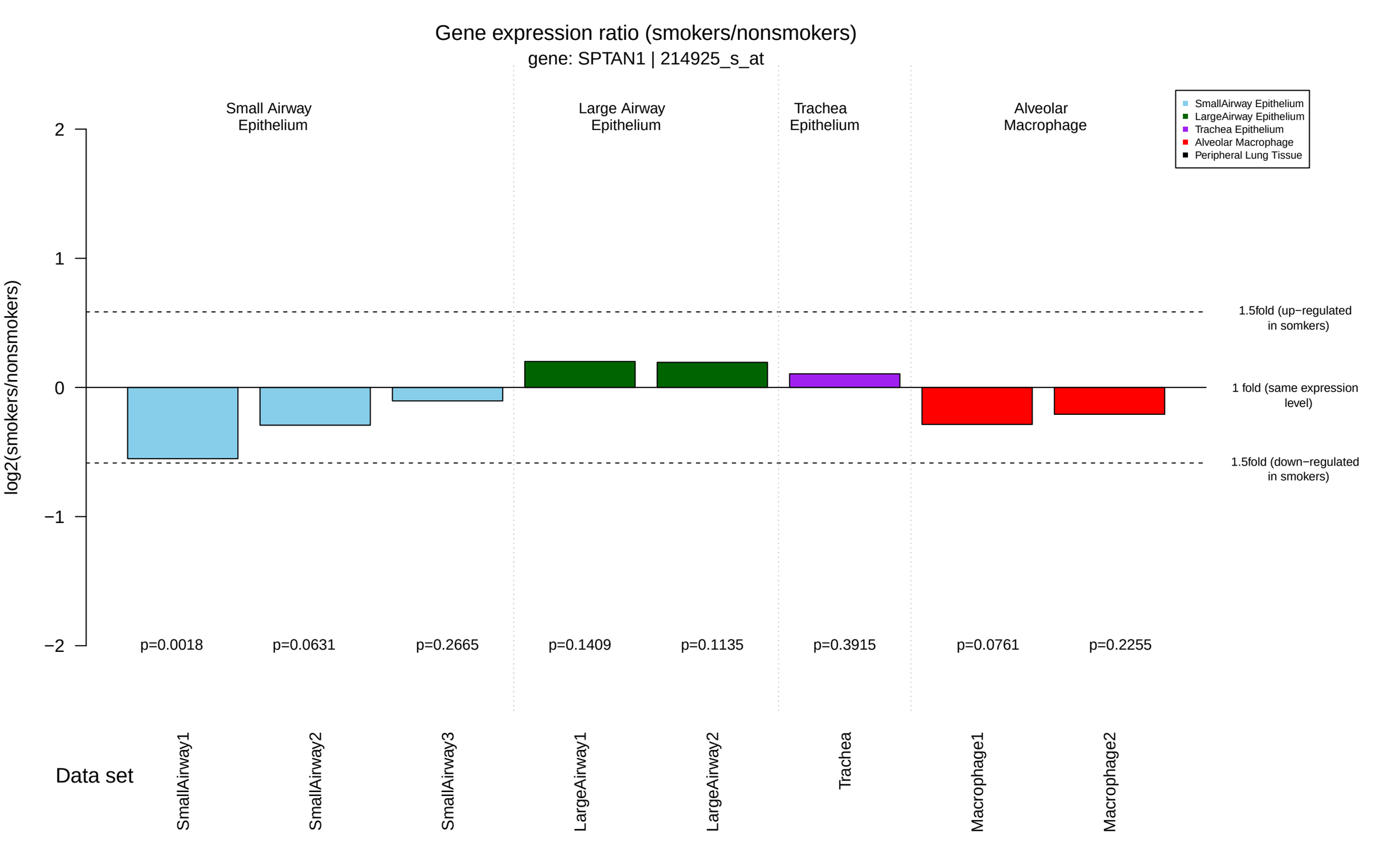
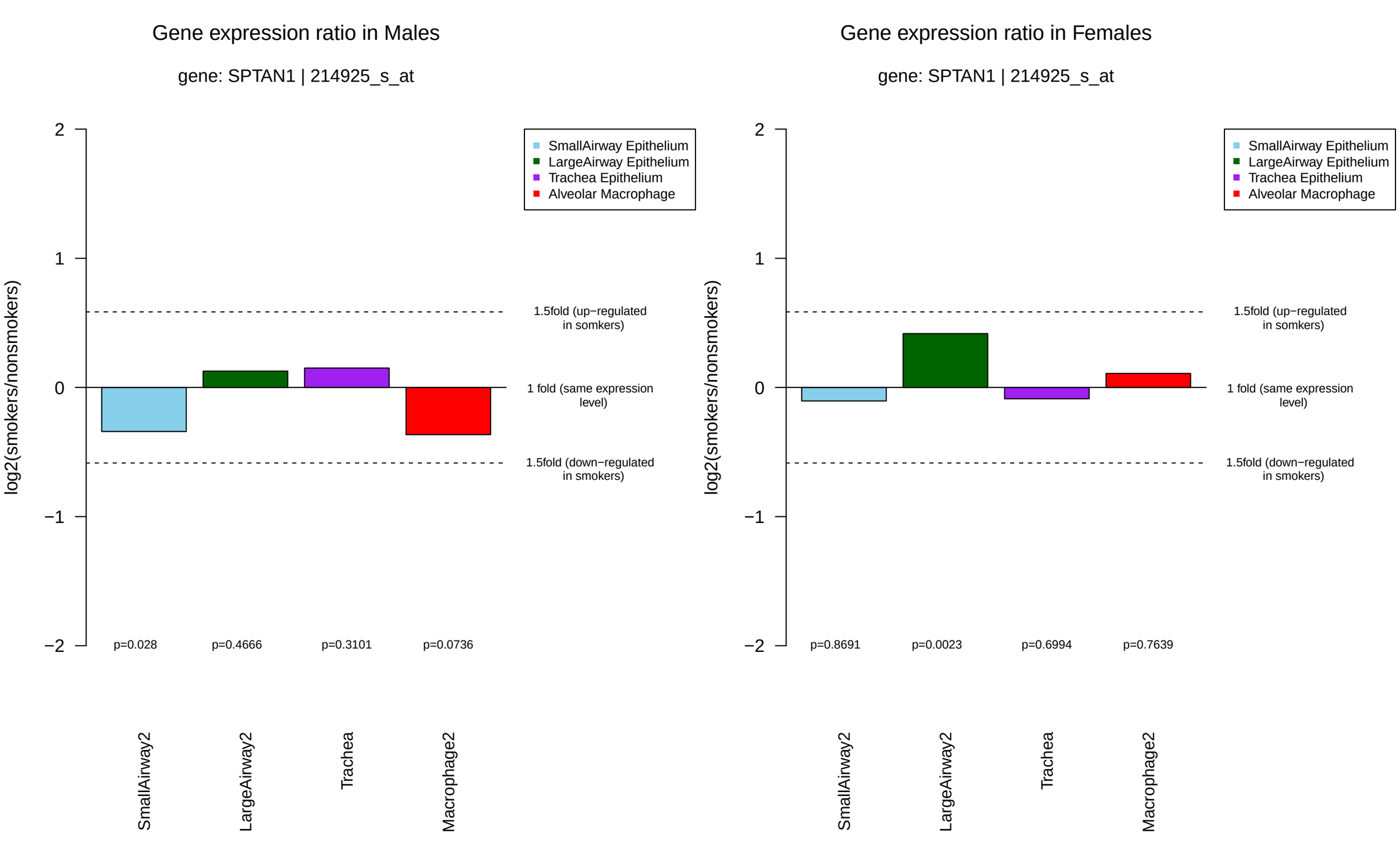
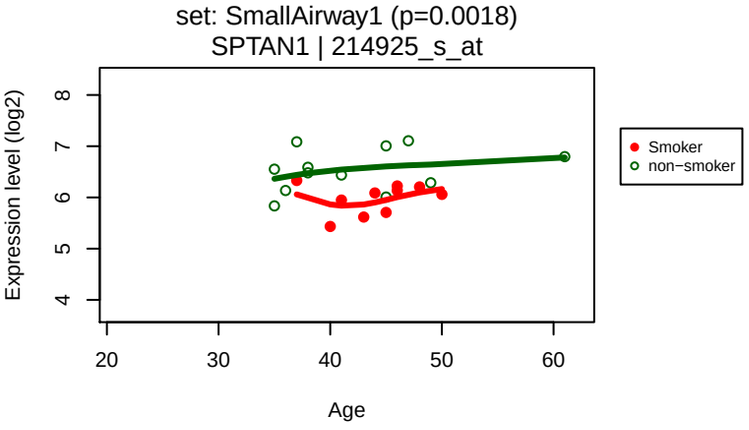
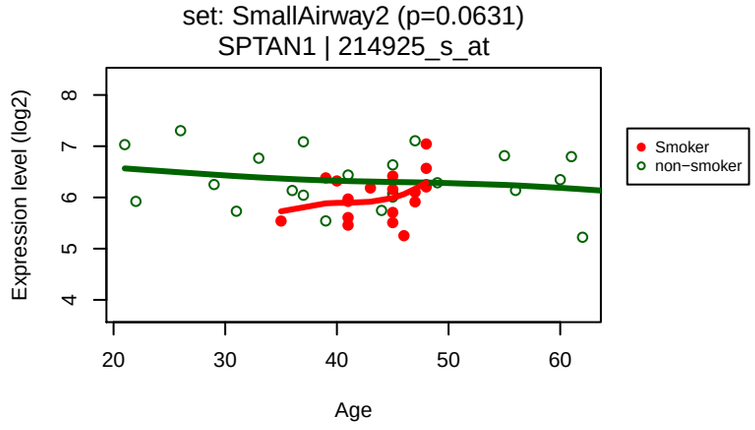
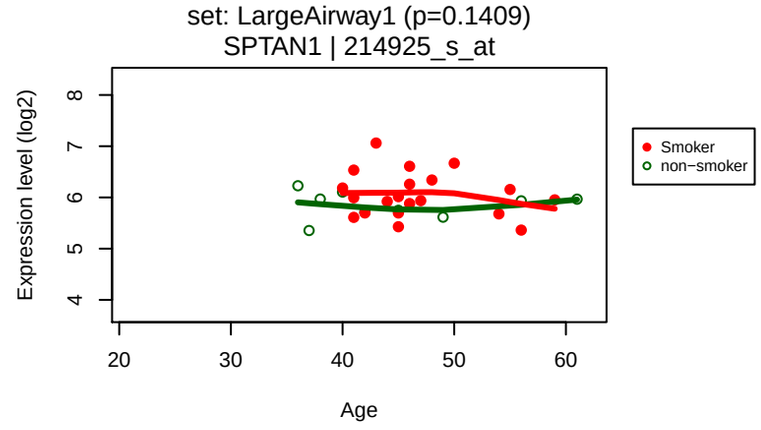

Gene expression of smokers and nonsmokers
gene: SPTAN1 | 214925_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0018
p=0.0631
p=0.2665
p=0.1409
p=0.1135
p=0.3915
p=0.0761
p=0.2255
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

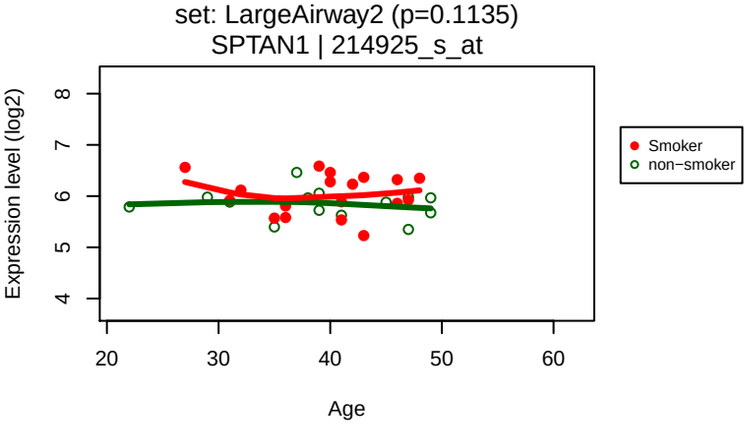

Gene expression ratio (smokers/nonsmokers)
gene: SPTAN1 | 214925_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0018
p=0.0631
p=0.2665
p=0.1409
p=0.1135
p=0.3915
p=0.0761
p=0.2255
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

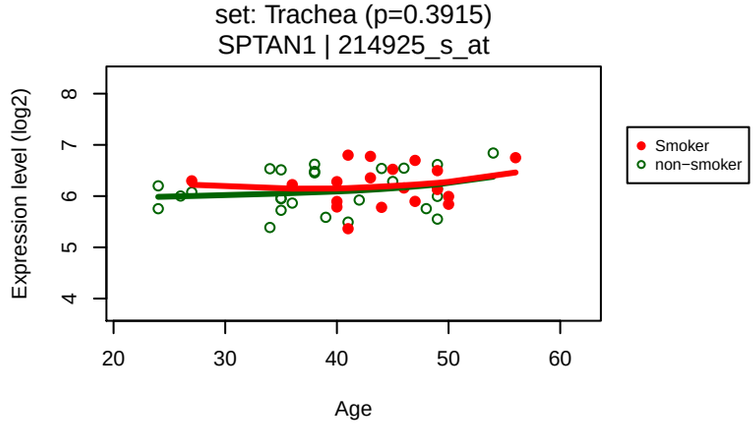

Gene expression ratio in Males
gene: SPTAN1 | 214925_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.028
p=0.4666
p=0.3101
p=0.0736
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPTAN1 | 214925_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8691
p=0.0023
p=0.6994
p=0.7639
SmallAirway2
LargeAirway2
Trachea
Macrophage2

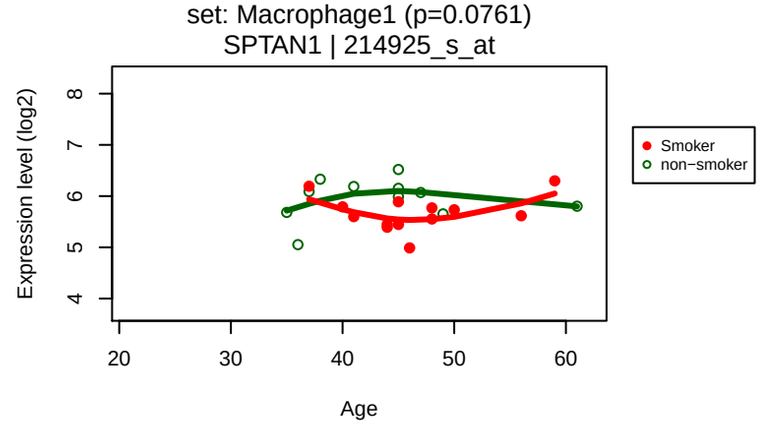

set: SmallAirway1 (p=0.0018)
SPTAN1 | 214925_s_at
Expression level (log2)
Age
Smoker
non−smoker

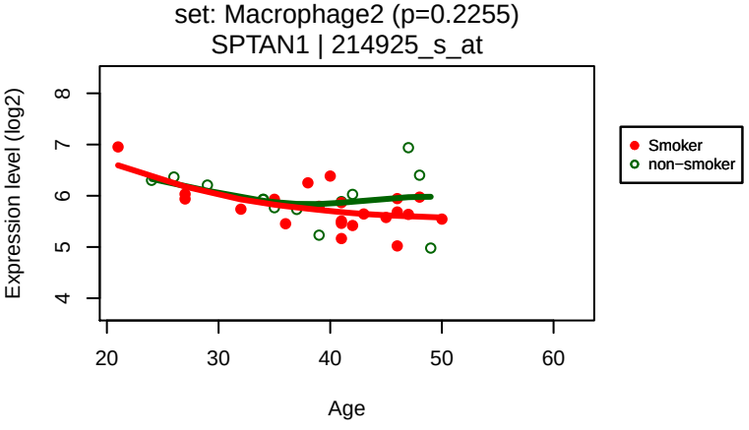

set: SmallAirway2 (p=0.0631)
SPTAN1 | 214925_s_at
Expression level (log2)
Age
Smoker
non−smoker

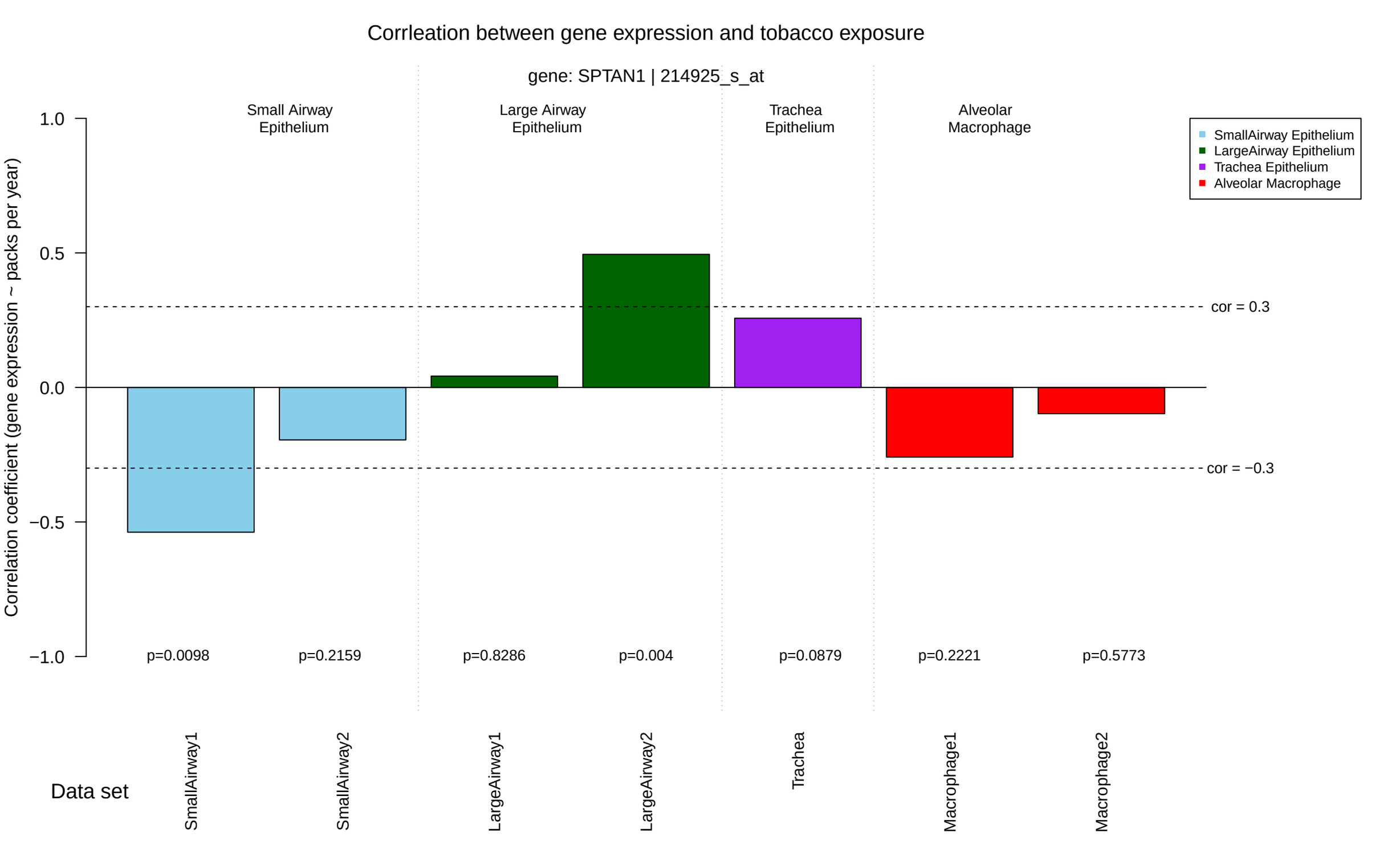

set: LargeAirway1 (p=0.1409)
SPTAN1 | 214925_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1135)
SPTAN1 | 214925_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3915)
SPTAN1 | 214925_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0761)
SPTAN1 | 214925_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2255)
SPTAN1 | 214925_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPTAN1 | 214925_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0098
p=0.2159
p=0.8286
p=0.004
p=0.0879
p=0.2221
p=0.5773
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2